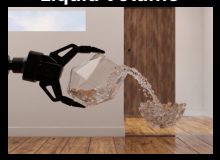
Physical Structure

Liquid Volume

Which container has larger volume?

Deformable

Is that a peach or a mug under the cloth?

Rigid Containment

How to fit all toys into the containers?

Specular Reflection

Reflection Authoring

Is the plant In the mirror or real?

Spatial Relations

Which object is closer to the observation position?

Correspondence

Which bag corresponds to the reflected one?

Enumerative Perception

Counting w Occlusion

How many air purifiers are there?

Spatial Segmentation

How many kettles are there in this scene?

Merged Observation

How many apples are there in this scene?

Category Ambiguity

How many sandwiches are there?

Structural Enclosure

How many balls are there in this scene?

Illumination Variability

How many fillets are there in this scene?

Perceptual Grounding

Partial Occlusion

Is that a golf stick or an umbrella behind the wall?

It is an umbrella.

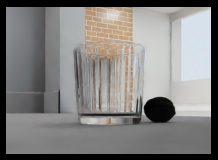
View Hallucination

Is there a cabinet there?

No, It's a piano

Material Transparency

Is the chestnut inside the glass?


No, it's outside.

Cognitive Mapping

Connectivity

Is the living room and the kitchen connected?

Regional Boundary

Is the sink and the chair located in the same area?

Traversable Passage

I need to navigate from the bedroom to the living room. Do I have to pass through the kitchen?

Yes, you do.

Long-Term Navigation

I need to navigate from the living room to the kitchen. Which room should I go through first?

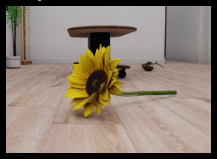
The piano room.

Metric Comparison

Dimensional Size

Which vase is larger?

The one under the table.

Spatial Distance

Which flower is closer to the table?

The darker one.

Physical Dynamics

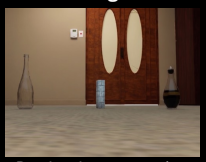
Inclined Plane

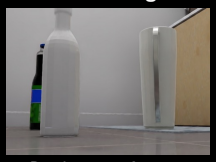
Can the apple rest on the slope?

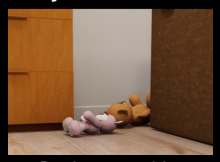
Stacking & Stability

How to stack and stabilize the three objects?

Spatial Relations

Linear Alignment

Do the three containers form a line?

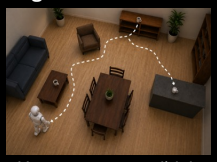
Geometric Configuration

Do the cups form an equilateral triangle?

Physical Contact

Do the two teddy bears touch?

Temporal Understanding

Unobserved Change

Is there any change inside the closed box?

Agent Observation

How many mugs did the other robot pass by?

Action Sequencing

Action Order Inference

What's the action sequence given spatial constraints?